

Notes: Recurrence Relation

1. What is a Recurrence Relation?

A recurrence relation is an equation that defines each term of a sequence as a function of its previous terms. It expresses the n th term of a sequence in terms of earlier terms.

Example: $a(n) = a(n-1) + 2$ means every term is obtained by adding 2 to the previous term.

2. General Form of a Recurrence Relation

General form: $a(n) = f(a(n-1), a(n-2), \dots, a(n-k))$

where $a(n)$ = n th term, f = some function, k = order of the recurrence.

3. Solution of a Recurrence Relation

A solution is a formula that gives the value of $a(n)$ directly without recursion.

Example: $a(n) = a(n-1) + 2$ with $a(0) = 1$ has solution $a(n) = 2n + 1$.

4. Method 1: Iteration (Substitution) Method

Steps:

1. Write the recurrence and initial condition.

Example: $a(n) = a(n-1) + 2$, $a(0) = 1$.

2. Expand step by step:

$$a(n) = a(n-1) + 2 = (a(n-2) + 2) + 2 = a(n-2) + 4 \dots$$

$$\text{After } k \text{ steps: } a(n) = a(n-k) + 2k$$

3. Stop when you reach the initial condition: $a(0) = 1$.

$$\text{Thus, } a(n) = 1 + 2n.$$

Example (Iteration Method)

Recurrence: $a(n) = a(n-1) + 5$, $a(0) = 4$

Solution: $a(n) = 4 + 5n$.

Recurrence: $a(n) = 2a(n-1)$, $a(0) = 1$

Solution: $a(n) = 2^n$.

5. Method 2: Characteristic Equation Method

Used for linear recurrence relations with constant coefficients.

General form: $a(n) = c_1 \cdot a(n-1) + c_2 \cdot a(n-2) + \dots + c_k \cdot a(n-k)$.

Steps:

1. Write recurrence in standard form.

2. Form the characteristic polynomial by assuming $a(n) = r^n$.

3. Solve for r (roots of the polynomial).

4. Write the general solution depending on type of roots.

5. Use initial conditions to solve for constants.

Case 1: Distinct Roots

If roots are distinct $r_1, r_2, \dots, r_k$ then
 $a(n) = A_1 \cdot r_1^n + A_2 \cdot r_2^n + \dots + A_k \cdot r_k^n$.

Example: $a(n) = 3a(n-1) + 4a(n-2)$, $a(0)=1$, $a(1)=7$.
Characteristic equation: $r^2 - 3r - 4 = 0 \rightarrow$ roots: 4, -1.
General solution: $a(n) = A \cdot 4^n + B \cdot (-1)^n$.
Using conditions $\rightarrow A=8/5$, $B=-3/5$.
Final solution: $a(n) = (8/5) \cdot 4^n - (3/5) \cdot (-1)^n$.

Case 2: Repeated Roots

If a root r has multiplicity m , the solution has the form:
 $a(n) = (B_1 + B_2 \cdot n + B_3 \cdot n^2 + \dots + B_m \cdot n^{(m-1)}) \cdot r^n$.

Example: $a(n) = 6a(n-1) - 9a(n-2)$, $a(0)=2$, $a(1)=12$.
Characteristic: $r^2 - 6r + 9 = 0 \rightarrow$ root $r=3$ (multiplicity 2).
Solution: $a(n) = (A + Bn) \cdot 3^n$.
Using conditions: $A=2$, $B=2$.
Final solution: $a(n) = (2 + 2n) \cdot 3^n$.

Case 3: Complex Roots

If roots are complex: $r = \alpha \pm i\beta$, then solution is expressed using sines and cosines:
 $a(n) = (C_1 \cdot \cos(n\theta) + C_2 \cdot \sin(n\theta)) \cdot \alpha^n$, where $\theta = \arctan(\beta/\alpha)$.

This ensures real solutions when the recurrence has real coefficients.

6. Applications of Recurrence Relations

1. Algorithm Analysis (e.g., Merge Sort: $T(n) = 2T(n/2) + O(n)$).
2. Counting Problems (Fibonacci sequence).
3. Dynamic Programming.
4. Computer Graphics (fractals, recursive images).
5. Mathematical Modeling (population growth, forecasting, etc.).

Summary

- Recurrence relation defines terms using previous ones.
- Two solving methods: Iteration and Characteristic Equation.
- Distinct roots $\rightarrow$ sum of terms $A \cdot r^n$.
- Repeated roots $\rightarrow$ multiply by powers of n (like $(A+Bn) \cdot r^n$).
- Complex roots $\rightarrow$ use sine and cosine forms.
- Applications: algorithms, Fibonacci, DP, modeling.